The Kervarie Invariant One Problem

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1. Exotic Spheres
2. A Geometric Definition of the Kervaire Invariant
3. The Kervaire Invariant One Problem
4. The Last Case in Dimension 126

Exotic Spheres

Surgery exact sequence for spheres:

$$\cdots \to \mathscr{S}(\mathbb{S}^n) \to [\mathbb{S}^n, G/O] \to L_n(\pi_1(\mathbb{S}^n)) \to \cdots$$

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- $\Theta_n := \mathscr{S}(\mathbb{S}^n)$ is the group of homotopy n-spheres under connected sum.
- $L_n(\pi_1(\mathbb{S}^n)) = L_n(0)$ is 4-periodic.

$$L_n(0) = \begin{cases} \mathbb{Z}, & n = 4k, \\ \mathbb{Z}/2, & n = 4k+2, \\ 0, & n \text{ odd}. \end{cases}$$

Exotic Spheres

- $[\mathbb{S}^n, G/O] = \pi_n(G/O)$.
  G/O fits into the fibration sequence $O \to G \to G/O$, which induces a long exact sequence of homotopy groups:

$$\cdots \to \pi_n(O) \xrightarrow{J_n} \pi_n(G) \cong \pi_n^{st}(\mathbb{S}) \to \pi_n(G/O) \to \cdots .$$

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  This splits into short exact sequences:

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  Moreover,

$$\mathrm{Ker}(J_{n-1}) \cong \begin{cases} \mathbb{Z}, & n = 4k, \\ 0, & \text{otherwise.} \end{cases}$$

Exotic Spheres

Theorem (Kervaire-Milnor, 1963)

Suppose $n \geqslant 5$. Let $\Theta_n^{bp} := \mathrm{Ker}(\Theta_n \to \pi_n(G/O))$.

(1) For $n \neq 4k+2$, there is a short exact sequence

$$0 \to \Theta_n^{bp} \to \Theta_n \to \pi_n^{st}(\mathbb{S})/J \to 0.$$

(2) For $n = 4k+2$, there is an exact sequence

$$0 \to \Theta_n^{bp} \to \Theta_n \to \pi_n^{st}(\mathbb{S})/J \xrightarrow{\Phi_n} \mathbb{Z}/2 \to \Theta_{n-1}^{bp} \to 0.$$

Φ_n: the Kervaire invariant.

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Φ_n: the Kervaire invariant.

Question:The Kervaire Invariant One Problem (algebraic version)

Is Φ_n surjective for a given dimension n?

Exotic Spheres

Theorem (Kervaire-Milnor, 1963)

For $n \geqslant 5$, the subgroup Θ_n^{bp} is cyclic,

$$|\Theta_n^{bp}| = \begin{cases} 1, & n \text{ even}, \\ 1 \text{ or } 2, & n = 4k+1, \\ 2^{2k-2}(2^{2k-1}-1)c_k, & n = 4k+3. \end{cases}$$

c_k: the numerator of $\frac{4B_{2k}}{k}$
B_{2k}: Bernoulli number.

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Therefore, the computation of Θ_n was transformed into the followings:

- computation of $\pi_n^{st}(\mathbb{S})/J$,
- the extension problem, and
- the Kervaire invariant one problem.

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Framed Manifolds

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Definition

A stably framed n-manifold is an equivalence class of triples $(M, i : M \hookrightarrow \mathbb{R}^{n+N}, t : \nu(M) \xrightarrow{\cong} M \times \mathbb{R}^N)$, where

- M is a closed smooth n-manifold,
- $i : M \hookrightarrow \mathbb{R}^{n+N}$ is an embedding,
- $t : \nu(M) \xrightarrow{\cong} M \times \mathbb{R}^N$ is a trivialization of the normal bundle,

under the equivalence relation generated by $(M, i, t) \simeq (M, i', t')$ for

$$i' : M \hookrightarrow \mathbb{R}^{n+N} \hookrightarrow \mathbb{R}^{n+N+1}, t' : \nu'(M) \cong \nu(M) \oplus \varepsilon \xrightarrow[\cong]{t \oplus \mathrm{id}} M \times \mathbb{R}^{N+1}$$

A Geometric Definition of the Kervaire Invariant

Let (M, i, t) be a stably framed $2m$-manifold.
For any $x \in H^m(M; \mathbb{Z}/2) \cong [M_+, K(\mathbb{Z}/2, m)]$, the following composition

$$\mathbb{S}^{2m+N} \to \mathrm{Th}(\nu(M)) \xrightarrow[\cong]{t} \mathrm{Th}(M \times \mathbb{R}^N) \cong \Sigma^N M_+ \xrightarrow{\Sigma^N x} \Sigma^N K(\mathbb{Z}/2, m)$$

determines an element $\phi(x) \in \pi_{2m+N}(\Sigma^N K(\mathbb{Z}/2, m))$.

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A calculation shows that $\pi_{2m+N}(\Sigma^N K(\mathbb{Z}/2, m)) \cong \mathbb{Z}/2$ for N large enough. Therfore, this defines a map of sets $\phi : H^m(M; \mathbb{Z}/2) \to \mathbb{Z}/2$.

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Proposition

The map ϕ is a quadratic refinement of the cup product on $H^m(M; \mathbb{Z}/2)$, in the sence that $\phi(x + y) = \phi(x) + \phi(y) + (x \cup y)$.

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Definition (Browder, 1969)

The Kervaire invariant $\Phi(M) \in \mathbb{Z}/2$ of (M, i, t) is the Arf invariant of the quadratic form ϕ; that is, $\Phi(M) = 1$ iff $|\phi^{-1}(1)| > |\phi^{-1}(0)|$.

Properties of the Kervaire Invariant

- Framed cobordant manifolds have the same Kervaire invariant, so it induces a map $\Phi : \pi^{st}_{2m}(\mathbb{S}) \to \mathbb{Z}/2$.

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- For $\psi \in O(N)$ and $(M, i : M \hookrightarrow \mathbb{R}^{2m+N}, t)$ a stably framed $2m$-manifold, we have

$$\Phi(M, i, t) = \Phi(M, i, \psi \circ t : \nu(M) \xrightarrow{t} M \times \mathbb{R}^N \xrightarrow{\mathrm{id}_M \times \psi} M \times \mathbb{R}^N).$$

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In the stable homotopy group of spheres, this orthogonal group action is captured by the J-homomorphism $J : \pi_*(O) \to \pi_*^{st}(\mathbb{S})$, so Φ factors through $\pi_{2m}^{st}(\mathbb{S})/J$.

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- In dimention $4k+2$, the Kervaire invariant $\Phi : \pi^{st}_{4k+2}(\mathbb{S})/J \to \mathbb{Z}/2$ coincides with the surgery obstruction. In particular, a framed $(4k+2)$-manifold can be converted into a homotopy sphere via surgery iff it has Kervaire invariant 0.

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- The Kervaire invariant is trivial in dimension $4k$.

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- $n = 126$ (Lin-Wang-Xu, 2024).

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Adams Spectral Sequences

Theorem (Browder, 1969)

There exists a stably framed n-manifold with Kervaire invariant one iff

- $n = 2^{j+1} - 2$,
- and in the Adams spectral sequence of sphere, the class $h_j^2 \in E_2$ survives to E_∞-page.

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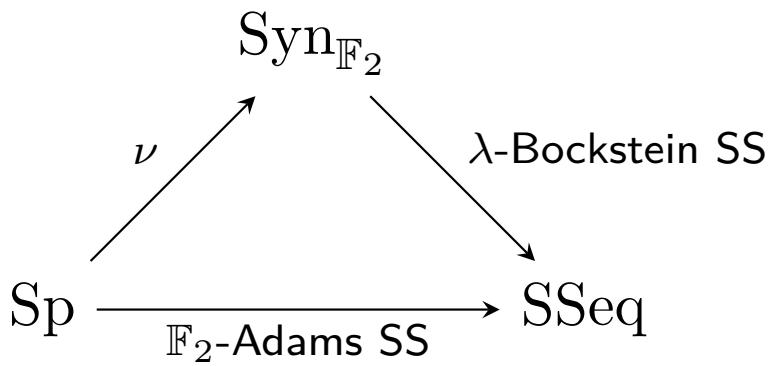
- $\mathbb{F}_2$-based Adams spectral sequence of sphere:

$$E_2^{s,t} = \mathrm{Ext}_{A_*}^{s,t}(\mathbb{F}_2, \mathbb{F}_2) \implies \pi_{t-s}(\mathbb{S}_2^\wedge)$$

- (E_2, d_2) is purely algebraic (Baues-Jibladze, 2006; Baues, 2006; Nassau, 2012), and the homotopy data in $\mathbb{S}$ is encoded in the higher differentials d_r.

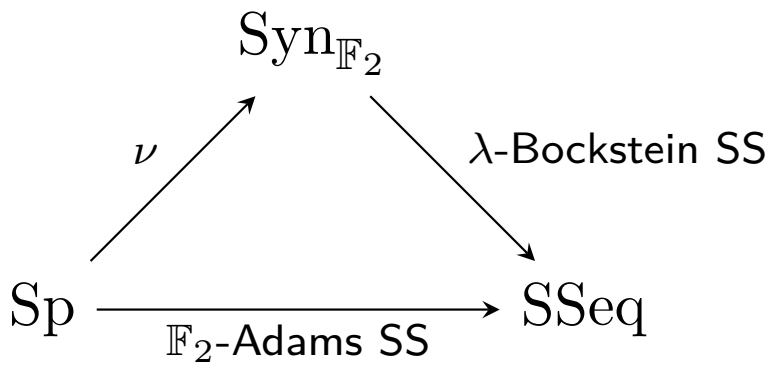
Synthetic Spectra

- In 2018, Pstragowski defined the stable ∞-category $\mathrm{Syn}_{\mathbb{F}_2}$ of $\mathbb{F}_2$-synthetic spectra, which categorifies the $\mathbb{F}_2$-based Adams spectral sequences.



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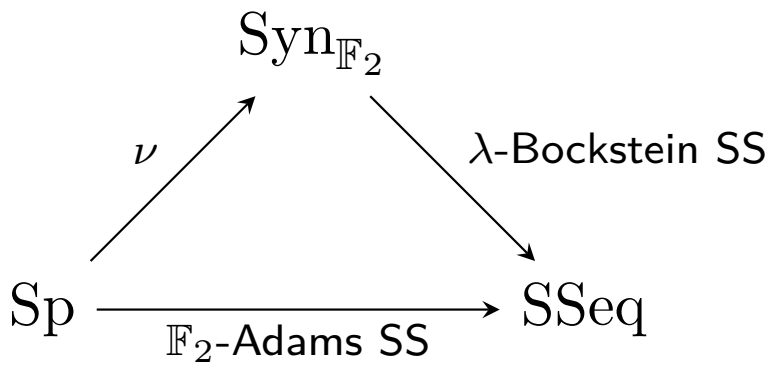
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- Applying the theory of synthetic spectra, Lin-Wang-Xu introduced two computational mathods: the Generalized Leibniz Rule and the Generalized Mahowald Trick.
- Based on these theoretical developments, Lin made computer programs that compute (E_2, d_2) and deduce new differentials (using the two methods) in a very effective way.

The Last Kervarie Invariant One Problem

Theorem (Lin-Wang-Xu, 2024)

The element h_6^2 survives to the E_∞-page in the Adams spectral sequence.

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- For degree reasons, h_6^2 cannot be killed by any differential in the Adams spectral sequence of sphere.
- There are 105 additive generators in stem 125 of the classical Adams E_2-page - these are the potential targets that could be hit by a nonzero differential from h_6^2.

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- There are 105 additive generators in stem 125 of the classical Adams E_2-page - these are the potential targets that could be hit by a nonzero differential from h_6^2.
- Using Lin's program and the inductive approach, thay can rule out 101 out of the 105 elements.
- They use adhoc arguments to prove that the remained 4 cases cannot happen and conclude that h_6^2 survives in the Adams spectral sequence.